

Lecture 11: Hardware Technologies for Stereoscopic Displays

“How does a VR headset know where your head is pointing?”

“How does HoloLens place holograms exactly on your table?”

Explain:

The magic is not only in displays, but in **hardware + sensors + tracking systems**.

VR Headsets Overview

Introduce three major categories:

1. Oculus / Meta Quest (Consumer VR)

- Standalone headset
- Inside-out tracking
- Controllers + hand tracking
- Used for:
 - Gaming
 - Education
 - Virtual classrooms

2. HTC Vive (High-end VR)

- PC-powered VR
- External sensors (Lighthouse tracking)
- Very high accuracy
- Used for:
 - Simulation
 - Research labs
 - Engineering training

3. Microsoft HoloLens (AR/MR Device)

- See-through AR headset
- Projects holograms on real world

- Spatial mapping
 - Used for:
 - o Medical visualization
 - o Industrial maintenance
 - o Architecture
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Key Hardware Components in Headsets

Explain common components inside headsets:

Component	Function
Displays	One screen per eye / split display
Lenses	Increase field of view
GPU / Processor	Real-time rendering
Cameras	Track environment
IMU (gyro + accel)	Motion detection
Depth sensors	Measure distance
Microphones	Voice input
Speakers	Spatial audio

Explain simply:

Without sensors, VR would just be a blind helmet.

Sensors and Tracking Systems (Core Technical Part)

Types of sensors:

1. IMU Sensors

(IMU = Inertial Measurement Unit)

- Accelerometer → movement
- Gyroscope → rotation
- Magnetometer → direction

Used for:

- Head rotation
- Controller motion

2. Camera-based Tracking

- Inside-out tracking (Quest, HoloLens)
- Cameras detect:
 - Room features
 - Hands
 - Controllers

Advantage:

- No external sensors needed
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3. External Tracking Systems

(Used in Vive)

- External base stations emit signals
 - Headset detects timing → calculates position
 - Extremely accurate
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4. Depth Sensors

Used in HoloLens:

- Measure distance to objects
 - Enable spatial mapping
 - Allow virtual objects to sit on real surfaces
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Why Tracking Matters for Stereoscopic Experience

Explain consequences:

If tracking is bad:

- Scene moves late
- Objects drift
- Eye–hand mismatch

- Motion sickness

If tracking is good:

- Realistic immersion
- Natural interaction
- Stable 3D perception

Explain:

Stereoscopic display gives depth, but tracking gives **presence**.

Recap + Quick Questions

Recap:

- Oculus = consumer standalone VR
- Vive = high accuracy VR
- HoloLens = AR/MR
- Sensors (IMU, cameras, depth) enable tracking
- Good tracking = good immersion

Quick oral questions:

- What is inside-out tracking?
- Which device uses spatial mapping?
- Why is IMU important in VR headsets?